Md Sakib Anwar

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Research Statement

My research develops principled, scalable program-analysis techniques to secure modern software, with a focus on logical flaws in memory-safe languages. I have exposed the disconnect between security research and developers and built practical tools that both block attacks and reveal gaps in vulnerability reports. By advancing symbolic execution, constraint solving, and inductive invariant synthesis, I detect and prove logical and resource-exhaustion vulnerabilities that still drive real-world breaches despite advances in type and memory safety. I aim to keep creating practical defenses against real threats while centering both end users and developers.

Education

The Ohio State University, PhD in Computer Science and Engineering – USA — 2026

- Advancing Logical Vulnerability Assessment via Detection & Input Synthesis.
- Advised by Dr Zhiqiang Lin and Dr Carter Yagemann.

The Ohio State University, MS in Computer Science – USA — 2024

- Major: Security & Privacy.
- Minors: Software Engineering & Programming Languages, Software Systems.

University of Dhaka, BSc in Computer Science – Bangladesh — 2019

- Undergraduate research in wireless localization and embedded systems.
- Leadership in IEEE student activities.

Publications

GoSonar: Detecting Logical Vulnerabilities in Memory Safe Languages Using Inductive Constraint Reasoning — 2025
Md Sakib Anwar, Carter Yagemann, Zhiqiang Lin
IEEE Symposium on Security & Privacy (S&P)

VerDiff: Vulnerability Presence Verification for Comprehensive Reporting Using Constraint Programming — 2025
Md Sakib Anwar, Carter Yagemann, Zhiqiang Lin
Annual Computer Security Applications Conference (ACSAC)

Extracting Threat Intelligence From Cheat Binaries For Anti-Cheating — 2023
Md Sakib Anwar, Chaoshun Zuo, Carter Yagemann, Zhiqiang Lin
International Symposium on Research in Attacks, Intrusions and Defenses (RAID)

A Comparative Study on GPR-based Indoor Positioning Systems — 2018
Md Sakib Anwar, Fariha Hossain, Nusrat Mehajabin, et al.
Intl. Conference on Innovations in Science, Engineering and Technology (ICIET)

Manuscripts in Preparation

GoSym: Semantic-Rich Symbolic Execution for Detecting Resource Exhaustion Vulnerabilities in Go
Md Sakib Anwar, Carter Yagemann, Zhiqiang Lin
Work in Progress

Presents a semantic-rich symbolic execution engine for Go, made with Go. Enables accurate representation of runtime behaviour facilitating efficient detection of ellusive vulnerabilities.

GoReX: Automated Resource Exhaustion by Synthesizing Constraint-Based Inductive Invariant
Md Sakib Anwar, Carter Yagemann, Zhiqiang Lin
Work in Progress

Introduces an symbolic execution guided inductive invariant generation technique to trigger nth-step resource exhaustion or loop escalation. It builds on the idea of loop invariant and how the constraints can be formalized in building a triggering input.

Teaching Experience

Graduate Teaching Assistant, The Ohio State University – USA — SP'21, AU'21

Teaching assistant for introductory programming and computational problem-solving courses.

- CSE 1222: Conducted lab sections for >50 students.
- Conducted office hours, evaluated assignments/exams, developed assessment materials.
- Managed graders and collaborated with professors to address student performance and disciplinary matters.
- Rated highly by students for clarity and approachability.

Guest Lecturer, The Ohio State University – USA — AU'25

Guest lecture for an undergraduate security projects course.

- Presented VerDiff (ACSAC'25) in CSE 5472: Information Security Projects.
- Adapted research using real-world analogies and toy examples.
- Focused on high-level security intuition and conducted a live demo.

Mentoring Experience

Summer Research Mentor, The Ohio State University – USA — SU'22, SU'25

Mentored undergraduate researchers in systems security and program analysis.

- Supervised projects on reverse engineering, program analysis, and OSS security analysis.
- Guided student in analyzing unity game and engineering a cheat.
- Mentored student in analysis of OSS security and construction of a dataset.
- Designed training modules for static and dynamic program analysis techniques.

Industry Experience

Software Engineer, KONA Software Lab Limited – Bangladesh — 2019 – 2020

Backend developer for large-scale digital financial platforms.

- Built secure transaction APIs and backend logic.
- Collaborated on service optimization and system architecture improvements.

Academic Service

External Reviewer: USENIX Security, IEEE S&P, NDSS, CCS, RAID, ACSAC (2021–2025)

Journal Reviewer: ACM Computing Surveys (2025)

Community & Leadership: Treasurer, IEEE Computer Society Student Branch Chapter, University of Dhaka (2018–2019)

Grants and Awards

Student Travel Grant Recipient, IEEE S&P — 2025

Student Travel Grant Recipient, ACSAC — 2025

Patents & Disclosures

VerDiff: Automated Vulnerability Version Detection for Open Source Security — 2024

Provisional Patent, The Ohio State University, Tech Disclosure T2024-006